Chapter 32 Review

Campbell Biology, AP Edition, 12e

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Summary of Key Concepts

Do organisms with radial symmetry, such as sea anemones, have *Hox* genes? Explain your thinking. **(BIG IDEAS 1 & 3)**

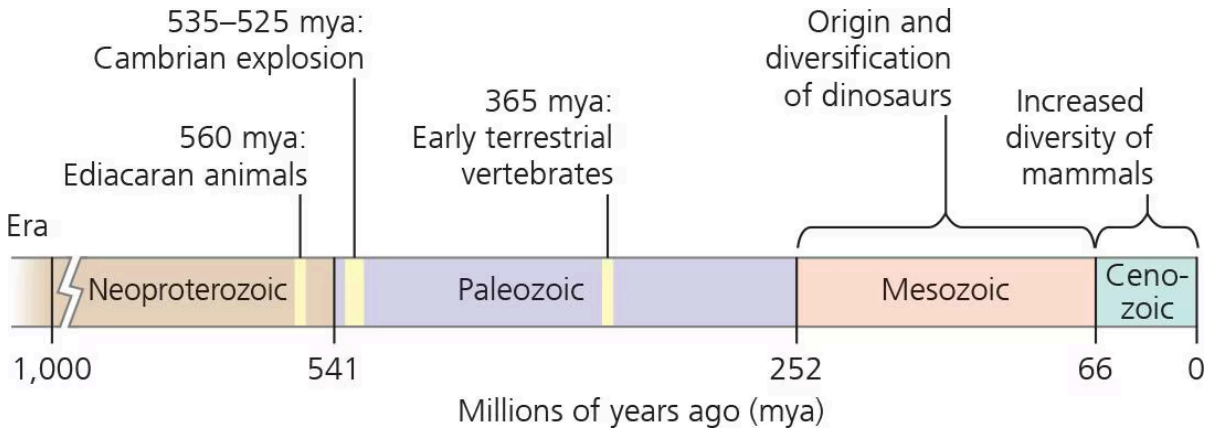
Concept 32.1: Animals are multicellular, heterotrophic eukaryotes with tissues that develop from embryonic layers

- Animals are heterotrophs that ingest their food.
- Animals are multicellular eukaryotes. Their cells are supported and connected to one another by collagen and other structural proteins located outside the cell membrane. Nervous tissue and muscle tissue are key animal features.
- In most animals, ****__gastrulation__ follows the formation of the ****__blastula__ and leads to the formation of embryonic tissue layers. Most animals have *Hox* genes that regulate the development of body form. Although *Hox* genes have been highly conserved over the course of evolution, they can produce a wide diversity of animal morphology.

Describe key ways that animals differ from plants and fungi.

Concept 32.2: The history of animals spans more than half a billion years

- Fossil biochemical evidence and molecular clock analyses indicate that animals arose over 700 million years ago.
- Genomic analyses suggest that key steps in the origin of animals involved new ways of using proteins that were encoded by genes found in choanoflagellates.



From left to right, Neoproterozoic Era from 1000 to 541 million years ago. 560 million years ago: Ediacaran animals; 535 to 525 million years ago; Cambrian explosion. Paleozoic Era from 541 to 252 million years ago; 365 million years ago: Early land animals. Mesozoic Era from 252 to 66 million years ago (origin and diversification of dinosaurs). Cenozoic Era from 66 million years ago to present (increased diversity of mammals).

What caused the Cambrian explosion? Describe current hypotheses.

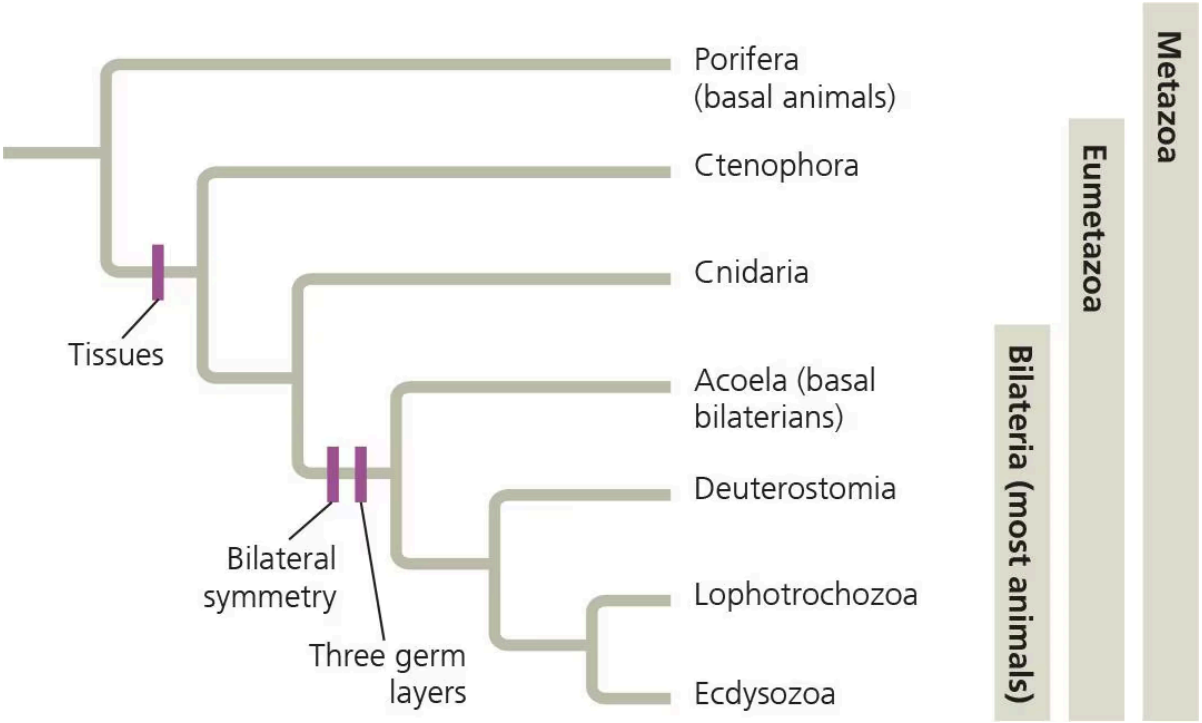
Concept 32.3: Animals can be characterized by body plans

- Animals may lack symmetry or may have **radial** or **bilateral symmetry**. Bilaterally symmetrical animals have **dorsal** and **ventral** sides, as well as **head** and **tail** ends.
- Eumetazoan embryos may be ****__diploblastic__ (two germ layers) or **triploblastic** (three germ layers). Triploblastic animals with a ****__body cavity__ may have a **coelom** or a **hemocoel** (or both).
- **Protostome** and **deuterostome development** often differ in patterns of ****__cleavage__, coelom formation, and ****__blastopore__ fate.

Describe how body plans provide useful information yet should be interpreted cautiously as evidence of evolutionary relationships.

Concept 32.4: Views of animal phylogeny continue to be shaped by new molecular and morphological data

- This phylogenetic tree shows key steps in animal evolution:



Phylogenetic tree with key steps in animal evolution. Tree is described as a list.

Branch Point 1 (Metazoa): Porifera (basal animals).

Branch Point 2, Tissues (Eumetazoa): Ctenophora

Branch Point 3: Cnidaria.

Branch Point 4, bilateral symmetry and three germ layers (Bilateria (most animals)): Acoela (basal bilaterians).

Branch Point 5. Deuterostomia.

Branch Point 6. Lophotrochozoa. Ecdysozoa.

Consider clades Bilateria, Lophotrochozoa, Metazoa, Chordata, Ecdysozoa, Eumetazoa, and Deuterostomia. List the clades to which humans belong in order from the most to the least inclusive clade.